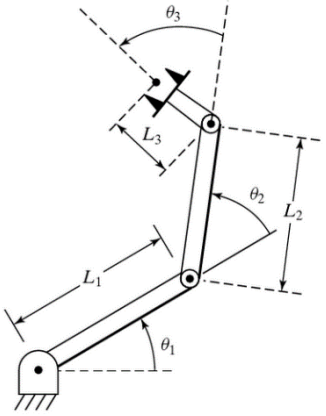
ENN 581 – Semester 2 – 2026
Week 9 Practical on dynamics and control

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Consider the 3-link planar manipulator shown below.



The geometry of the robot is given as L1 = 3m, L2 = 2m, L3 = 1m. The robot starts from rest where the end effector pose is given as (5,1,90^o). We would like to move it to (-3,3, 180°) and come to a rest there.

In week 6, you learnt how to determine the trajectory. Say you are using a cubic interpolation between joint angles.
a. Compute the trajectory of each joint and plot it.

In week 8, you learnt how to compute the joint torques at each point in the trajectory. This is the inverse dynamics problem. The masses are given as m1 = 3kg, m2 = 2kg, m3 = 1kg. Assume that they are concentrated at the distal ends of each link.
b. Use the Lagrangian formulation to obtain an equation of motion of the following form.

$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q}) + \boldsymbol{f}(\mathbf{q}, \dot{\mathbf{q}})$$

Now, let us try to develop a controller for the robot. For the sake of simplicity, consider the first link at the first instance of time. Ignore friction.

c. In the initial configuration, compute the matrices M, c and g.
d. The desired position q_d is known. At each timestep, it is the next position that the joint must move to.
e. Write down a partitioned control law for the joint torque.
f. Give this torque as input to the system (i.e., write a program that takes torque as input, follows the system dynamics, and gives joint positions and velocities as outputs. In other words, write a forward dynamics simulator – recall from the lecture that forward dynamics is useful for simulation and inverse dynamics is useful for control).
g. Repeat this at multiple instances of time along the trajectory and plot the error in position as a function of time.
h. Change the mass in your control law to an imprecise value like 2.8 kg. Observe the evolution of error over time.
i. Can you extend the controller to all the joints?